

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 19)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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5	
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7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Table 1 shows the energy inputs and food energy outputs of three farming systems.

Table 1

System	Energy input / GJ ha ⁻¹ yr ⁻¹	Food energy output / GJ ha ⁻¹ yr ⁻¹
Extensive cereal farming (low inputs)	4	20
Intensive cereal farming (fertiliser, pesticide, machinery)	20	60
Intensive beef (grain-fed)	40	4

0 1

. 1

Calculate the energy ratio (output / input) for each system.

[2 marks]

Answer

0 1

. 2

Use your answers to explain the difference between productivity and efficiency.

[3 marks]

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0 1

. 3

Explain what is meant by an **energy subsidy** and give **three** examples.

[3 marks]

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01 . 4

Explain why spreading the same total inputs over a larger area would increase total food production.

[2 marks]

10

02 . 1

Explain how each of the following new technologies improves the validity of ecological research.

[8 marks]

Satellite / radio / GPS tracking

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DNA databases and eDNA

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Image recognition software

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Acoustic monitoring / sonograms

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0 2 . 2

Explain why new technologies allow the collection of 'more representative' data.

[2 marks]

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10

0 3

Table 1 shows some of the IUCN criteria used to place species in Red List categories.

Table 1

Category	Population decline over 10 years or 3 generations	Area of habitat	Number of mature adults	Probability of extinction in the wild
Vulnerable	over 40%	under 2000 km ²	under 10 000	over 10% in 100 years
Endangered	over 60%	under 500 km ²	under 2500	over 20% in 20 years or 5 generations
Critically endangered	over 80%	under 10 km ²	under 250	over 50% in 10 years or 3 generations

0 3**1**

State **three** roles of the IUCN.

[3 marks]

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0 3**2**

A species of lemur has declined from 12 000 to 2 100 mature adults in three generations and occupies 400 km² of forest. Use Table 1 to determine its Red List category. Explain your answer.

[3 marks]

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0 3**3**

Explain why only a small proportion of species have been assessed for the Red List, and give **two** criteria used to decide which species to assess.

[3 marks]

03 . 4

Give **one** example of a species whose Red List category improved because of conservation and state the actions responsible.

[1 mark]

10

04 . 1

Explain how overfishing of the giant triton mollusc damaged coral reefs in Australia.

[2 marks]

04 . 2

Describe **four** management practices used in marine protected areas to protect coral reefs.

[4 marks]

1

2

3

4

0 4 . 3

Explain why deep-water coral reefs recover much more slowly from trawling damage than tropical reefs.

[2 marks]

0 4 . 4

Give **two** threats to deep-water reefs and **one** conservation measure.

[2 marks]

10

0 5 . 1

Explain how each of the following reduces the environmental impact of fishing.

[8 marks]

Square-mesh escape panels

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Minimum and maximum catchable sizes

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No-take zones

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Closed seasons

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0 5 . 2

Explain why catch quotas work well for pelagic species such as mackerel but poorly in mixed demersal fisheries.

[2 marks]

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10

06 . 1

Describe coppicing and explain how it increases the biodiversity of a woodland.

[4 marks]

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06 . 2

Explain how the reintroduction of grey wolves to Yellowstone National Park increased biodiversity.

[4 marks]

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06 . 3

Explain why culling or removing species may be necessary in habitat management.

[2 marks]

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07 Table 1 compares organic and inorganic fertilisers.

07 . **1** Complete Table 1 by giving two advantages and two disadvantages of each type of fertiliser.

[6 marks]

Table 1

Fertiliser	Advantages	Disadvantages
Organic (manure, sewage sludge, compost) (eg)		
Inorganic (eg ammonium nitrate, NPK)		

07 . **2** Explain how hydroponic cultivation maximises productivity and give **two** disadvantages.

[4 marks]

10

08 . 1

Give **two** habitats created by farming that support important wildlife communities.

[2 marks]

1

2

08 . 2

Explain how drainage and nutrient enrichment of farmland reduce biodiversity.

[2 marks]

08 . 3

Explain **one** problem that can be caused by introducing a biological control species.

[1 mark]

5

0 9 Table 1 lists some feedback mechanisms that affect the Earth's climate.

0 9 . **1** Complete Table 1 by stating whether each mechanism is an example of **negative** or **positive** feedback.

[4 marks]

Table 1

Mechanism	Type of feedback	
Higher temperatures increase cloud cover, raising albedo		
Melting permafrost releases methane		
Higher CO ₂ levels increase photosynthesis and carbon sequestration		
Melting ice reduces albedo		

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0 9 . **2** Explain why natural systems regulated by negative feedback rarely reach tipping points.

[2 marks]

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0 9 . **3** Using an example, explain how a large unusual event can cause a tipping point.

[2 marks]

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09 . 4

Explain how the human response to negative feedback in agriculture can make food production less sustainable.

[2 marks]

10

10 . 1

Explain how an understanding of ecological succession and population dynamics can inform conservation decision-making.

[9 marks]

Extra space

10 . **2** State what is meant by a **taxon**.

[1 mark]

Do not write outside the box

10

Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

OR

1 1 . 2

Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

Do not write outside the box

END OF QUESTIONS